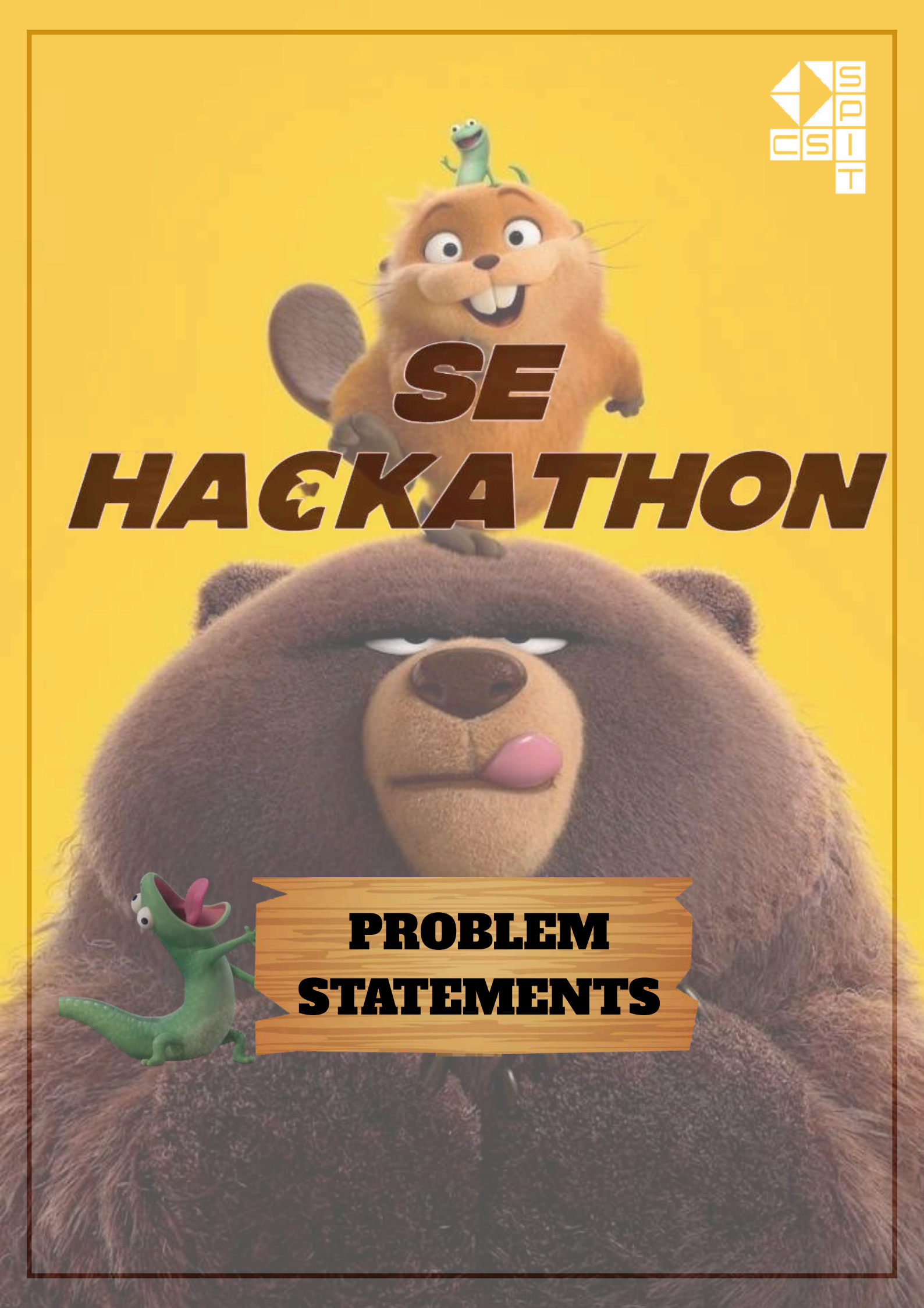
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WEB DEV

PS 1

NEXUS: THE AUTOMATED CLEARANCE PROTOCOL

Hritani is a final-year student who should be celebrating her upcoming graduation. Instead, she is caught in a bureaucratic nightmare. She spends her afternoons running between the library, laboratory, and accounts office, hunting for physical signatures on a "No Dues" paper that is easily lost. She needs a centralized, transparent system that handles the heavy lifting of administrative clearance so she can focus on her future. Nexus is a digital-first protocol designed to transform the chaotic graduation exit process into a seamless online experience.

MAIN FEATURES:

Multi-Stage Approval Workflow: The core engine of Nexus. It replaces physical signatures with a digital "Chain of Custody." When Hritani submits her documents, the system triggers a sequential notification system: first to the Departmental Lab In-charge, then the HOD, and finally the Principal. Each authority has a dedicated "Action Dashboard" to approve or flag applications with comments.

Smart Document Vault with Verification: Students upload scanned copies of their ID cards, library receipts, and lab manuals. The system uses basic metadata checking to ensure files are in the correct format (PDF/JPEG) and within size limits. Admins can view these side-by-side with the approval buttons for rapid verification.

Digital Certificate Generator: Upon final approval from the Principal, the system uses a server-side PDF generator to produce a finalized, non-editable "Digital No-Dues Certificate" and a consolidated Transcript, ready for immediate download.

Internal Dues Reconciliation Engine: Instead of complex external APIs, Nexus uses a "Flat-File Upload" system for departments. The Librarian can upload a CSV of students with pending books, and Nexus automatically flags those students' profiles, blocking their clearance until the status is updated to "Paid" via the admin panel.

Integrated Payment Sandbox: A simulation-ready payment layer. For any detected dues, such as library fines or hostel repairs, Nexus integrates a payment gateway. Once a transaction is successful, the system generates a digital receipt and automatically moves the student to the next stage of the clearance pipeline.

Live Status Heatmap: A real-time progress tracker for the student. It provides a color-coded visual map showing which departments have cleared them (Green), which are pending (Yellow), and where there is a "No Dues" rejection (Red), eliminating the need for manual inquiries.

BROWNIE POINTS:

QR-Coded Verification: Every certificate generated contains a unique QR code. When scanned, it leads to a simple public URL on the project site that confirms the certificate was officially issued to the student for their specific batch, proving the document is genuine.

Automated Email Nudge System: A script that runs every 24 hours to check for "stale" applications. If an authority has not looked at a request for two days, Nexus sends an automated, professional email reminder to their inbox to keep the process moving.

Personal Digital Locker Export: A "One-Click Download" feature that bundles all of the student's uploaded documents, receipts, and final certificates into a single, organized ZIP file, serving as a permanent digital archive for their career records.

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WEB DEV

PS 2

SHELFLIFE: THE RESURRECTION OF THE LIVING ARCHIVE

Rayan and Siddh are an engineering duo drowning in the "Digital Graveyard." Rayan has 400 open tabs across three windows, while Siddh has thousands of bookmarks that are essentially a list of broken redirects. Between the chaos of their research and the speed of the modern web, their shared knowledge is dying faster than they can read it. They need more than a list: they need a living, breathing ecosystem that organizes itself, summarizes the noise, and physically decays if neglected. SHELFLIFE is a social, AI-augmented curation engine designed for teams like them, rewarding obsessive discovery and punishing digital hoarding through a "Biological" visual interface.

MAIN FEATURES:

Autonomous Ingestion Pipeline (The Scraper): A sophisticated background worker architecture. When a URL is dropped into their shared space, a headless browser auto-fetches the title, metadata, and a full-page screenshot. This data is fed into a GenAI model to generate a 3-sentence executive summary, ensuring neither Rayan nor Siddh ever has to manually label a link again.

Semantic Vibe-Engine (GenAI Tagging): Using LLM-based sentiment analysis, every saved link is auto-classified into "Energy Types" (e.g., Chaotic, Educational, Cursed, or High-Signal). These are visualized as "Mood Pills" that adapt their color and glow intensity based on the content's sentiment, creating a multi-sensory browsing experience for the whole team.

Automated Curator Archetypes: A data-driven profiling system. Based on the "Vibe Distribution" of their saves, the system assigns Rayan and Siddh dynamic "Curator Cards" as shareable identities.

Biological Decay and Compost System: A unique "Visual Aging" algorithm. Links that remain unclicked for 14 days begin to desaturate and shrink. After 30 days of neglect, the link "dies" and is moved to a global, public "Compost Heap" (a graveyard page). This creates a sense of collective urgency for the duo to engage with their saved knowledge.

Real-time Collaborative Synapse: A high-concurrency layer powered by WebSockets. Rayan and Siddh can see each other's live cursors, real-time link additions, and instant emoji reactions. The "Shelf Weather" visualizes their combined activity: a shelf looks "Stormy" during heavy sessions and "Foggy" when abandoned.

Perplexity Search-Grounding: Every saved link features a live "Context Feed" powered by search-grounded AI. It tracks how the topic has evolved since the day it was saved, providing Rayan and Siddh with a real-time update on whether the content is still relevant or if a more recent "successor" exists.

BROWNIE POINTS:

Redis Distributed Synapse: A high-speed memory layer acting as the system's central brain. It synchronizes real-time telemetry, such as cursor paths and live decay signals, across multiple server nodes. This ensures that Rayan and Siddh experience a singular, perfectly aligned shelf reality regardless of which server instance is hosting their individual sessions.

Remix and Lineage Architecture: A "Git-style" versioning system for curation. Other teams can fork Rayan and Siddh's public shelves, creating a new branch. The system tracks the "Lineage Tree," showing the original curators and how many times their collection has been remixed.

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APP DEV

PS 1

LUMINA: THE PROACTIVE ENGINEERING SIDEKICK

Renee is an engineering student who feels like she is constantly chasing her own schedule. Between the chaos of college activities, hidden deadlines in emails, and a rigid attendance system, she needs an assistant that does not just store data but predicts her needs. Lumina is a proactive agent designed for the offline, high-stress reality of engineering life.

MAIN FEATURES:

AI Timetable Parser and Attendance Engine: The heart of Lumina's scheduling. User simply uploads a PDF or image of his timetable. Lumina uses on-device OCR to learn his schedule automatically. It tracks attendance in real-time and provides Bunk Analytics, calculating exactly how many more classes he can miss to stay above the 75% threshold.

ContextSwitch: "You said you were studying. Your phone disagrees." Lumina monitors foreground app changes and screen events to build a real-time task-switching timeline. It calculates a Cognitive Debt Score using an exponential decay model, penalizing rapid context switches. Users can join "Study Squads" to share anonymized flow graphs, creating social pressure to maintain deep focus.

Group Discussion Hub : A real-time collaboration suite. A persistent group chat powered by Socket.io, featuring Whiteboard and Pasteboard utilities. Renee can drop a code snippet or sketch a logic gate into the chat, and it stays pinned for the whole group to edit live.

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Gmail and Calendar Heatmap Sync: A visual Stress Meter that pulls from Google Calendar and Gmail. It scans for academic keywords in emails to auto-populate a color-coded heatmap, turning the UI Red when deadlines collide and Green when it is time to relax.

Multi-User Kanban & Progress Engine: A shared task board for team projects. It ensures everyone in the group is aligned on project phases. Renee can easily categorize tasks into What I am Doing and What I Want to Do, bridging the gap between goals and daily execution.

Frictionless Expense Logger: A rapid-entry tool for daily spends. It generates a Weekly Student Wrap, helping Renee manage her budget through simple, non-intrusive CRUD operations.

Local-First Second Brain (RAG): User can query his textbook PDFs and notes entirely offline. Using a local vector database (TensorFlow.js), Lumina performs Semantic RAG. Data never leaves the device, ensuring total privacy even in signal-dead zones like college basements.

BROWNIE POINTS:

Overleaf/LaTeX Remote Preview: An integration that allows Vatsal to view the latest compiled version of his group's LaTeX lab report directly within the Lumina dashboard for quick mobile reviews.

Smart Battery Guardian: If the phone battery drops below 15%, the agent proactively pauses heavy background RAG indexing and pushes a Final Sync of the Kanban so the team has his latest updates before the phone dies.

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APP DEV

PS 2

AETHER: THE AUTONOMOUS CAMPUS OPERATING SYSTEM

Priyank and Harshav are an engineering duo exhausted by the friction of daily campus life. Priyank identified that the university functions as a collection of disconnected silos: communication is buried in fragmented groups, and resource scheduling is trapped in a loop of physical signatures. Harshav realized that the lack of a common digital operating layer creates a massive cognitive load for both students and faculty. Together, they decided to build Aether: a modular, role-based mobile platform that unifies academic, administrative, and social workflows into a single, high-integrity interface.

MAIN FEATURES:

Role-Aware Intelligent Mobile Dashboard: The primary entry point for every user. Priyank sees his next class location and club alerts, while a Professor sees real-time attendance trends and pending leave approvals. Ensure t dashboard widgets and push notifications are synchronized across all server nodes, providing zero-latency updates to the mobile client.

Smart Mobile Workflow and Approvals Engine: A digital "Chain of Responsibility" for campus processes like room bookings or certificate requests. The engine supports multi-stage approvals, secure digital authentication, and automated document generation for instant viewing. If Harshav submits a request on his phone, the system identifies the correct HOD and Principal for immediate notification and one-tap approval.

AI Campus Copilot : A context-aware mobile assistant that guides users through campus life. Harshav can ask natural language questions about scholarships or pending tasks via text or voice and receive specific, actionable steps derived from live workflow data.

Faculty Coordination Workspace: A specialized toolset for professors to manage student advising, attendance-related actions, and departmental collaboration on the go. It reduces context-switching by allowing faculty to handle routine coordination, notice publishing, and student follow-ups directly from their mobile device.

Conflict-Aware Scheduling Layer: A unified mobile calendar that merges academic timetables with operational events. It features a built-in "Clash Detection" algorithm and cross-calendar synchronization to prevent double-booking. Priyank can view room availability in real-time with the system providing "Smart Suggestions" for the best time slots based on his academic schedule.

Analytics and Decision Intelligence: A high-level mobile dashboard for campus leadership that surfaces operational trends. It uses data from all other modules to identify bottlenecks in approvals or service response times, allowing administrators to move from reactive fixing to proactive management via mobile alerts.

Mobile Resolution and Support Protocol: A structured system for reporting campus issues like IT failures or facility maintenance. Users can use their phone's camera to upload evidence such as high-resolution photos stored in distributed object storage. Support teams receive prioritized tickets, while leadership receives a "Heatmap" of recurring hotspots.

BROWNIE POINTS:

Extensible "Super App" Architecture: A DevOps-focused modular foundation. The team built the platform to be "Plugin-Ready," allowing other student developers to build their own mini-apps: such as canteen trackers or research portals: via an API-first design that ensures the core mobile shell can grow without a redesign.

Financial Settlement Gateway: A secure clearinghouse integration that allows students to clear library fines, canteen bills, or lab dues via instant UPI or card payments.

SE HACKATHON

AI ML

PS 1

LEDGERSPY THE AIR-GAPPED FORENSIC SIDEKICK

Piyush, a young Chartered Accountant, walks into a high-stakes audit engagement with a strict warning from the senior partner: "One data leak and we can lose our license." With his Wi-Fi disabled and the pressure mounting, Piyush launches LedgerSpy on his laptop. While his teammates are still manually scrolling through thousands of Excel rows, LedgerSpy quietly analyzes the entire ledger. By lunch, Piyush uncovers a sophisticated circular trading network involving ghost vendors that had been draining the client's funds for over three years. The partner is stunned not just by the fraud, but by how a single junior auditor using only his laptop just saved the entire engagement.

MAIN FEATURES:

Digital Frequency Profiling: Detects if transaction amounts are being fabricated by humans rather than occurring naturally. The auditor sees a clean bar chart comparing the ledger's leading digits against the ideal Benford's Law distribution, with clear color highlights on significant deviations.

Fuzzy Entity Reconciliation: Finds hidden vendor collusion where names are slightly altered to bypass checks. The user gets an interactive similarity matrix showing matched pairs (e.g., "Stationery Ltd" vs "Stationery L.T.D") along with shared payment patterns and risk scores.

Anomaly & Outlier Detection: Highlights unusual transactions without needing labeled fraud data. Auditors can explore a dynamic scatter plot where red-highlighted outliers are clickable, revealing full transaction details and context on hover or click.

Bank Statement Reconciliation: Automates vouching by matching ledger entries against scanned bank statements. The interface shows a clean side-by-side view with color-coded rows: green for matched, yellow for partially matched, and red for missing evidence.

Relational Risk Mapping: Reveals hidden links between employees, vendors, and transactions. The auditor interacts with a glowing node-link graph where node size and glow intensity reflect risk level and centrality, making key gatekeepers in a fraud ring instantly visible.

Explainable Risk Insights: Removes the black-box problem so findings can be defended in court. For every flagged item, the user sees a simple breakdown chart showing which factors contributed most to the risk score (e.g., "65% due to unusual timing + high amount").

Local Audit Memo Assistant: Turns complex analysis into professional language. With one click, the auditor gets a well-drafted audit summary in standard accounting tone, ready to copy into reports.

Data Integrity Dashboard: Ensures the data is clean before analysis begins. The user sees a beautiful dashboard with a prominent "Readiness Score" percentage, along with categorized issue lists and visualizations for null values, duplicates, and format mismatches.

BROWNIE POINTS:

Monte Carlo Going Concern Stress Test: Simulates thousands of possible cash-flow futures. The auditor views an intuitive probability chart showing the chance of the company surviving the next 12 months, with scenario bands colored from safe (green) to critical (red).

Privacy-Preserving Industry Benchmarking: Lets the auditor check if the company's error or anomaly rate is normal for its sector.

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AI ML

PS 2

SENTIENCE: THE MULTIMODAL BEHAVIORAL INTELLIGENCE ENGINE

Aayush is a brilliant engineer with a flawless resume, but he consistently fails at the final hurdle: the high-pressure interview. He realizes that while he knows the technical answers, he struggles with the unspoken language of professional communication. His eye contact wavers when he is nervous, his posture collapses during difficult questions, and his tone lacks the confidence his words suggest. Aayush needs more than a resume checker: he needs a sophisticated behavioral mirror. Sentience is a multimodal AI/ML ecosystem designed to perform deep-inference analysis on not just what a candidate says, but how they exist in the digital space, providing a forensic breakdown of their interview performance.

MAIN FEATURES:

Kinematic Posture and Gesture Analysis Engine: Implement a computer vision system that extracts 3D skeletal coordinates to monitor Aayush's posture in real-time. The engine must calculate confidence metrics based on shoulder alignment and detect specific nervous gestures by analyzing the spatial relationship between hand and face landmarks over a moving temporal window.

Facial Expression and Vocal Sentiment Fusion: Develop a dual-stream processing pipeline that categorizes facial expressions into primary emotional states while simultaneously extracting vocal features such as pitch and energy.

Cross-Modal Semantic Validator: Create a validator that calculates the semantic similarity between Users's spoken answers and the claims made in his uploaded PDF resume. The system should flag incongruency spikes where his verbal explanation lacks the depth, technical keywords, or logical consistency present in his written history.

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Dynamic Agentic Interviewer: Build an intelligent agent that conducts the interview using adaptive logic. Instead of a static list of questions, the interviewer must utilize a decision-tree that pivots based on Aayush's real-time behavioral performance. If the system detects high stress or low confidence, the AI should change its questioning strategy to test his resilience.

Low-Latency Video Capture and Streaming SDK: Integrate a specialized video SDK to handle the real-time recording and streaming of the interview session. This must ensure high-resolution video is captured for ML inference without creating a bottleneck on the device's primary processor.

Real-Time Speech-to-Text Intelligence API: Utilize a speech-to-text integration to generate sub-second transcripts. This layer must provide the raw text input for semantic validation while providing word-level timestamps to map specific behavioral spikes to the exact words spoken.

Visual Engagement and Gaze Tracker: Develop a utility that calculates engagement scores by tracking pupil movement relative to the screen. By estimating head pose and gaze direction, the system must flag instances where Aayush loses eye contact with the virtual interviewer, indicating a drop in focus or confidence.

BROWNIE POINTS:

Persona-Driven Prompts: Allow Aayush to switch between different interviewer personalities, such as a "Supportive Mentor" or an "Aggressive Skeptic." Each persona features unique prompt-engineered behavioral triggers that alter the AI's tone, questioning pace, and reaction to the candidate's stress levels.

The "Devil's Advocate" Agent: This agent specifically targets moments of uncertainty to test Aayush's emotional stability, recovery speed, and ability to defend his technical logic under pressure.